

Supplementary Material: Selective Multi-Scale Transformer

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This supplement provides the complete per-pair audit and the source-only empirical gate details for the 17-pair annotated subset. All candidate selection and spatial consensus decisions are made without the hidden 3RScan alignment; matching uses every depth-valid sampled point, rescan observations are stored in native scan coordinates, and annotation labels are used only for post-hoc scoring and pose evaluation.

1 Full per-pair audit

The main paper reports weighted aggregates over 16 source pairs and one held-out target. Table 1 exposes every pair. “Sem.” is semantic precision among consensus inliers. The pose column uses the strict $5^\circ/0.25$ m gate.

The result is heterogeneous by construction. 4aca and 95be have near-zero retained semantic accuracy, but for different reasons: 4aca usually lacks a correct semantic candidate, whereas 95be often contains one but fails to recover it with consensus. 20c993 and pair07 have high retained semantic accuracy but very large pose error. The single strict source pose success is chairs, while the target is reported separately as a held-out point estimate.

2 Candidate-set availability audit

The evaluator optionally records whether any of the top-20 descriptor candidates has the query’s semantic or instance label. These fields are computed only after candidate construction and consensus, using annotations for audit purposes; they do not affect descriptor ranking, RANSAC, or the reported clean metrics. A correct semantic candidate is available for 79.7% of source queries and 96.6% of target queries. Only 25.8% and 63.6%, respectively, become correct semantic inliers; conditional recovery given a candidate hit is 32.4% on source and 65.8% on target.

The pair-level contrast is more informative than the aggregate. For 4aca, semantic candidate recall is 19.5% and correct-inlier coverage is 0.0%; this is a genuine candidate-set omission. For 95be, semantic candidate recall is 69.2%, but only 1.0% of candidate-hit queries become correct inliers, indicating wrong-consensus/pose selection rather than simple absence of the label. The audit also records 208 geometric inliers for 4aca whose candidate sets contain no correct semantic label, making explicit that a spatially coherent hypothesis is not automatically semantically valid.

3 Backbone scale control

The clean scale control uses the same depth-valid matching protocol, spatial consensus threshold, and RANSAC budget for ViT-L/14 and ViT-G/14. Table 3 reports the original 14-pair matched subset (13 source pairs plus the target) and the target separately. ViT-G gives a higher retained semantic-accuracy point estimate on the matched subset while the pose gate remains sparse, but the audit does not isolate a pre-consensus candidate-layer effect. A pair-bootstrap audit gives a +1.8 percentage-point weighted semantic difference with a 95% interval of $[-0.4, +5.3]$ points; the source-only 13-pair difference is +1.9 points with $[-0.6, +5.9]$. Both intervals include zero, so the scale-up result is descriptive rather than a claim of statistically established superiority.

4 Top-K sensitivity

We rerun the complete clean protocol over $K \in \{1, 5, 10, 20, 50\}$, keeping the ViT-G blocks (2, 20, 40), the 0.10 m consensus threshold, the RANSAC budget, and the source/target split fixed. The sweep is an audit of the fixed $K = 20$ operating point; no target result is used to choose K . On the source pairs, $K = 1$ has the highest weighted retained semantic accuracy but only 11.2% coverage. Among settings retaining at least 20% weighted source coverage, $K = 10$ is the most precise (89.3%), while $K = 20$ gives the larger coverage-oriented operating point used in the main paper (30.4%). The held-out target follows the same qualitative pattern up to $K = 20$; widening to $K = 50$ reduces retained semantic accuracy and loses the strict pose gate.

5 RANSAC seed sensitivity

The primary tables fix RANSAC seed 17 for exact reproducibility. To measure stochastic sensitivity without changing descriptors, candidates, threshold, or iteration budget, we rerun every pair with seeds 17, 29, and 41. The held-out target is stable: retained semantic accuracy ranges from 93.1% to 93.5% and coverage from 67.6% to 68.2%. The source aggregate is more sensitive, ranging from 83.1% to 88.0% retained semantic accuracy at 30.3–30.8% coverage; the variation is concentrated in wrong-pose pairs 95be, pair04, and pair06. Thus spatial consensus is a

Pair	Q	Inl.	Cov. (%)	Sem. (%)	Rot. (deg.)	Trans. (m)	Pose < 5°/0.25m
0cac	826	105	12.7	81.0	95.0	2.04	—
20c993	49	19	38.8	100.0	145.9	2.05	—
4aca	1000	227	22.7	0.0	159.3	1.22	—
5630	961	55	5.7	74.5	143.7	1.91	—
6bde	931	460	49.4	95.4	3.8	1.26	—
751a	1035	522	50.4	91.0	2.3	0.41	—
95be	1020	57	5.6	12.3	67.7	0.34	—
c670	840	204	24.3	87.7	5.8	0.34	—
chairs	1145	610	53.3	98.7	3.7	0.18	✓
pair02	1054	514	48.8	90.5	4.5	0.26	—
pair03	1181	295	25.0	92.2	60.5	0.31	—
pair04	1094	111	10.1	74.8	51.9	1.84	—
pair05	1170	688	58.8	93.8	12.3	0.23	—
pair06	1096	184	16.8	29.3	38.9	1.37	—
pair07	963	53	5.5	100.0	149.5	3.00	—
pair08	683	468	68.5	99.8	8.6	0.46	—
target	1069	727	68.0	93.5	1.5	0.17	✓

Table 1: Complete ViT-G/14 per-pair audit with $K = 20$, $\tau = 0.10$ m, and 1,500 RANSAC iterations. Q is the number of valid query patches and Inl. is the number retained by spatial consensus.

Split / pair	Cand. sem. recall (%)	Correct-inlier cov. (%)	Recovery cand. (%)	Miss inliers
Source weighted	79.7	25.8	32.4	368
Target held out	96.6	63.6	65.8	2
4aca	19.5	0.0	0.0	208
95be	69.2	0.7	1.0	10

Table 2: Post-hoc candidate-set availability audit at $K = 20$. “Recovery” is the fraction of queries with a correct semantic candidate that become correct semantic inliers. Miss inliers are geometric inliers with no correct semantic candidate in their top-20 set.

selective verifier, not a deterministic global-pose estimator on the hard source pairs. No seed was selected using target labels.

6 Descriptor-permutation negative control

To test whether spatial consensus alone could explain the semantic result, we repeat the exact ViT-G protocol

Setting	Cov. (%)	Sem. (%)	Rot. (deg.)	Pose < 5°/0.25m
ViT-L/14, original 14	33.8	85.1	—	2/14
ViT-G/14, original 14	34.3	86.9	—	2/14
ViT-L/14, target	65.1	92.4	3.97	1/1
ViT-G/14, target	68.0	93.5	1.54	1/1

Table 3: Clean ViT-L/14 to ViT-G/14 scale control.

after randomly permuting the reference descriptor rows while leaving reference points, labels, query points, $K = 20$, the 0.10 m threshold, and the 1,500-iteration budget unchanged. The permutation is fixed with seed 17 and is applied before candidate construction; it is not used to tune any threshold. Table 6 shows that the source aggregate falls to 46.3% retained semantic accuracy at 10.5% coverage, with no strict pose success. On the held-out target, coverage and retained semantic accuracy fall to 7.9% and 3.6%, respectively, with a 162.2° rotation error. This negative control supports the claim that the normal result depends on descriptor–point identity rather than on the RANSAC verifier alone; the residual semantic

K	Src. cov. (%)	Src. sem. (%)	Src. pose < 5°/0.25m	Tgt. cov. (%)	Tgt. sem. (%)	Tgt. pose < 5°/0.25m
1	11.2	89.5	2/16	27.9	92.3	1/1
5	20.7	88.8	2/16	51.6	92.8	1/1
10	25.3	89.3	0/16	60.8	93.4	1/1
20	30.4	85.0	1/16	68.0	93.5	1/1
50	37.3	78.5	3/16	58.7	90.0	0/1

Table 4: Clean top- K sensitivity. Source metrics are weighted over 16 pairs; target metrics are the held-out point estimate.

Split	Seed	Cov. (%)	Sem. (%)	Inst. (%)	Pose < 5°/0.25m
Source, 16 pairs	17	30.4	85.0	84.6	1/16
Source, 16 pairs	29	30.8	88.0	87.7	1/16
Source, 16 pairs	41	30.3	83.1	82.8	2/16
Target, held out	17	68.0	93.5	93.4	1/1
Target, held out	29	68.2	93.1	93.0	1/1
Target, held out	41	67.6	93.1	92.9	1/1

Table 5: RANSAC seed sensitivity with fixed candidates and budget.

floor on source pairs reflects category-level coincidences, not successful correspondence. At the paired scene level, normal matching is better on 15/16 pairs for both coverage and retained semantic accuracy; the macro semantic gap is 38.9 percentage points (76.3% versus 37.4%).

7 Frame-level selection

We separately test whether a lightweight frame-level confidence rule transfers across scenes. On the 16 source pairs, we select the feature/filter pair by macro retained semantic accuracy subject to 5% minimum coverage, and repeat the selection in leave-one-pair-out folds. The source-only winner is concatenated ViT-G layers (2, 20, 40) with reciprocal matching and a top-1/top-2 margin of 0.02; it is selected in 13/16 folds. We apply this fixed choice once to the held-out target without using target labels for selection. This is a retrieval confidence ablation, distinct from the main spatial-consensus result.

8 Source-only empirical gate fitting

The residual selector is fit on all 16 source pairs at a requested 90% retained semantic-accuracy target, then applied once to the target. We call this an empirical source-fit gate rather than a calibration guarantee. Table 8 also reports representative leave-one-pair-out (LOO) fits. LOO thresholds are fit without the listed pair, and the reported coverage and precision are on the held-out pair.

The all-source fit retains 236 of 1,069 target queries (22.1%) at 96.6% retained semantic accuracy. LOO gate fitting exposes the limitation of residual-only selection: held-out 4aca and 95be reach only 0.0% and 10.0% retained semantic accuracy, respectively. The candidate audit above shows that 4aca is candidate-set limited, while 95be has

Split	Coverage (%)	Semantic (%)	Strict pose < 5°/0.25m
Source, 16 pairs	10.5	46.3	0/16
Target, held out	7.9	3.6	—

Table 6: Descriptor-permutation negative control.

Split	Filter	Cov. (%)	Sem. (%)	RANSAC inlier	Rot. (deg.)
Source macro	one-way	100.0	62.5	13.1	53.56
Source macro	source-selected	11.3	68.6	24.0	49.69
Target	source-selected	12.1	87.6	48.1	2.86

Table 7: Source-only frame-level confidence selection. The target row uses the configuration selected on source pairs only.

substantial candidate availability but fails consensus recovery; neither is fixed by a threshold choice. Cross-fitted LOO pooling retains 1,364/15,048 source queries (9.1%) at 76.6% retained semantic accuracy; only 13/16 folds have a 95% threshold, pooling 115 queries at 85.2% accuracy. This is a transfer audit, not a universal guarantee. The threshold search scans residual prefixes and uses a one-sided empirical target; it does not provide a family-wise finite-sample guarantee.

The 0.03304 m source-fit threshold retains 6.1% of source queries with 91.7% observed accuracy and 14.1% of target queries at 98.7% accuracy and 97.1% semantic mIoU (Table 9). However, the threshold was selected after scanning 14,949 residual prefixes, so its raw 90.1% Clopper–Pearson lower bound is descriptive rather than family-wise valid. A Bonferroni correction over all prefixes produces no threshold satisfying a 90% bound; the maximum adjusted lower bound is 87.4%. For an auditable fixed point, a 0.02 m gate gives an unadjusted 90.9% source lower bound at 1.9% source coverage, and 3.3% target coverage at 100% retained semantic accuracy. Since this fixed point is also chosen retrospectively, it is an audit diagnostic, not a family-wise guarantee. The pair-bootstrap intervals and LOO failures remain the relevant evidence about scene heterogeneity.

9 Cross-scan label transfer

Table 10 reports the complete transfer table, including micro accuracy in the underlying CSV and class-balanced mIoU in the table.

At the target, consensus reaches 85.5% conditional semantic mIoU at 68.0% coverage. The source-fit selector reaches 89.5% semantic and 90.7% instance mIoU at 22.1% coverage. The source rows are retained to show that this utility is not uniform across the heterogeneous scene pool.

The conditional and abstention-aware numbers answer different questions. For example, the source-fit gate has 61.2% pooled conditional semantic mIoU but only 4.1% when rejected queries are counted as false negatives; the latter is the appropriate full-query utility diagnostic for a

Fit / held-out pair	Threshold (m)	Cov.	Sem. prec.	Inst. prec.
LOO 4aca	0.09606	21.6%	0.0%	0.0%
LOO 95be	0.04391	1.0%	10.0%	10.0%
LOO pair06	0.05469	5.9%	33.8%	33.8%
all source $\rightarrow$ target	0.04093	22.1%	96.6%	96.6%

Table 8: Empirical source-fit residual gate. The all-source row is evaluated on the target; LOO rows are evaluated on the held-out source pair.

Operating point	Split	Threshold (m)	Cov. (%)	Sem. (%)	LCB (%)
source-fit (raw)	source	0.03304	6.1	91.7	90.1
source-fit (raw)	target	0.03304	14.1	98.7	—
fixed 2 cm	source	0.02000	1.9	93.8	90.9
fixed 2 cm	target	0.02000	3.3	100.0	—

Table 9: Residual operating points. The source-fit row reports a raw post-selection lower bound; the fixed 2 cm row reports an unadjusted fixed-threshold query-level LCB. Neither is a family-wise guarantee after retrospective threshold inspection.

selective system.

As a data-quality check, an annotation-projection audit over all 32,835 sampled points finds a 2.74 cm median nearest-mesh distance, a 6.37 cm 95th percentile, and a 99.85% non-zero label rate. Only 0.31% of samples are farther than 0.30 m; these points are retained rather than silently filtered. The complete per-side audit is included with the released diagnostics.

10 Learned geometry control

To complement the classical FPFH control, we evaluate the published FCGF 3DMatch checkpoint with the maintained WarpConvNet adapter. This is a geometry-only learned baseline: it receives the sampled 3-D points and no RGB tokens. We use the same 14-pixel sampling, $K = 20$, 0.10 m spatial-consensus threshold, and hidden alignment only for evaluation. The 0.025 m voxel size is the FCGF extraction setting. Across all 17 pairs, top-20 FCGF reaches 18.7% weighted coverage and 53.3% weighted retained semantic accuracy, with 0/17 strict pose successes. The per-pair breakdown below makes its heterogeneity explicit.

Downstream geometry-control comparison. To test whether the correspondence comparison carries through to the concrete label-transfer use case, we rerun the same post-hoc semantic/instance transfer summary for the one-way FPFH and top-20 FCGF controls. Table 13 reports the consensus-inlier rows on the same 16-source/one-target split. ViT-G gives the highest source coverage and mIoU among the three controls. On the target, FCGF reaches 64.2% semantic mIoU at 17.5% coverage, while FPFH reaches 82.8% at only 13.2% coverage; FPFH is not an equal-density comparison because it voxelizes the sampled cloud before description. The query-level baseline diagnostics and exact aggregation are included in the artifact

Dataset / selector	Queries	Cov.	Sem. mIoU	Inst. mIoU
Source / all	15048	100.0%	19.4%	19.3%
Source / consensus	4572	30.4%	41.5%	39.0%
Source / source-fit	1339	8.9%	61.2%	51.5%
Target / all	1069	100.0%	67.4%	55.5%
Target / consensus	727	68.0%	85.5%	72.3%
Target / source-fit	236	22.1%	89.5%	90.7%

Table 10: Cross-scan semantic and instance label transfer. The reported mIoU is conditional and class-balanced over retained query patches; coverage is the fraction of valid query patches retained by the selector.

Split	Selector	Semantic mIoU (%)	Instance mIoU (%)
Source, 16 pairs	all	22.3	24.5
Source, 16 pairs	consensus	13.3	16.3
Source, 16 pairs	source-fit gate	4.1	5.0
Target, held out	all	67.4	55.5
Target, held out	consensus	55.4	48.8
Target, held out	source-fit gate	19.7	26.3

Table 11: Fixed-query, abstention-aware mIoU. The semantic and instance label universes are fixed from all non-zero query labels in each split; a rejected query predicts void and contributes a false negative, while zero-union classes receive IoU zero.

bundle.

11 Reproducibility and scope

Artifact boundary. The anonymized artifact release contains the paper sources, lightweight diagnostics, code, and consistency verifier; the venue-facing PDF package contains only the main paper and this supplement. Raw 3RScan frame archives and model checkpoints remain external inputs and are not silently bundled.

Protocol comparability. DUST3R, MAST3R, and VGGT are adjacent RGB foundation models whose standard outputs are pointmaps, image-level matches, or multi-view geometry; SemReg and ML-SemReg are trained semantic-registration systems. We do not report them as direct numeric baselines because porting those outputs to the saved RGB-D frame-bank, native-coordinate candidate pool, and post-hoc semantic label-transfer denominator would define a different experiment. FPFH and FCGF are therefore the controlled geometry-only baselines under the exact protocol used here, and no state-of-the-art registration claim is made. No MAST3R smoke result is included in this dated snapshot because a one-frame image-match experiment would not define the paper’s frame-bank or semantic-transfer denominator.

The numerical tables are generated from the saved diagnostics with:

```
conda run -n worldcache python \
  scripts/generate_supplementary_tables.py \
  --root REPO_ROOT \
  --output-dir REPO_ROOT/paper
```

Pair	Cov. (%)	Sem. (%)	Rot. (deg.)	Trans. (m)	Pose < 5°/0.25m
pair02	5.0	5.7	42.5	1.70	—
pair03	18.0	42.0	53.4	0.47	—
pair04	17.3	0.0	177.8	2.61	—
pair05	45.1	93.8	10.2	0.19	—
target	17.5	86.1	33.2	0.50	—
0cac	11.7	0.0	114.7	2.02	—
20c993	16.3	100.0	83.0	3.07	—
4aca	34.3	1.5	156.0	0.61	—
6bde	8.2	90.8	31.4	1.50	—
95be	12.5	0.0	143.8	2.73	—
5630	5.4	34.6	41.4	1.99	—
751a	15.1	82.1	12.0	0.44	—
c670	9.0	2.6	173.7	4.83	—
chairs	35.9	91.0	77.8	1.05	—
pair06	18.7	0.0	25.8	1.52	—
pair07	6.3	68.9	98.1	2.06	—
pair08	34.8	90.8	10.7	0.50	—

Table 12: Learned geometry-only FCGF control at top- $K = 20$.

Dataset / matcher	Cov. (%)	Sem. mIoU (%)	Inst. mIoU (%)	Sem. acc. (%)
Source / ViT-G	30.4	41.5	39.0	85.0
Source / FPFH	3.8	34.8	31.1	62.6
Source / FCGF	18.8	14.5	14.8	51.2
Target / ViT-G	68.0	85.5	72.3	93.5
Target / FPFH	13.2	82.8	82.8	90.8
Target / FCGF	17.5	64.2	64.2	86.1

Table 13: Downstream label-transfer control. All rows use spatial-consensus inliers; semantic and instance mIoU are class-balanced over retained queries. FPFH uses voxelized points and therefore has a different point denominator.

The final consistency audit is:

```
conda run -n worldcache python \
  scripts/verify_submission_consistency.py \
  --root REPO_ROOT
```